

15.5 Balanced and unbalanced one-way layouts

Every result so far holds for any group sizes, yet experimenters aim for balance. This section says what it buys: plain orthogonality, the largest guaranteed power for a fixed number of observations, and a test of equal means that tolerates unequal variances. The third matters most in practice, because unbalanced data with unequal variances are common and the classical F test can then go badly wrong.

15.5.1 Orthogonality and simplicity

In a balanced layout, orthogonality of contrasts is plain orthogonality $\sum_k c_k d_k = 0$ (Theorem 15.4.1(d)), so tabulated sets such as the integer polynomial contrasts can be used as they stand. The side conditions $\sum_k \alpha_k = 0$ and $\sum_k n_k \alpha_k = 0$ define the same parameters, the pooled variance is the plain average of the group variances, and every pairwise difference has the standard error $s\sqrt{2/m}$. Tukey's intervals have exact simultaneous coverage (Theorem 13.3.2), whereas with unequal sizes the Tukey–Kramer intervals (Eq. 13.3.3) are conservative by an unknown amount.

15.5.2 Balance maximizes guaranteed power

The noncentrality Eq. 15.3.3 depends on the group sizes as well as the means. No allocation of a fixed total is best for every configuration of means, and the planner does not know the means, so the relevant criterion is the power *guaranteed* against every configuration whose means span at least a range Δ , as in Proposition 11.5.2. For unequal group sizes, Exercise (B1) already identifies the least favourable configuration; what is new here is the comparison of allocations.

Proposition 15.5.1. Balance maximizes the guaranteed noncentrality

In a one-way layout with group sizes $n_1, \dots, n_g$, $g \geq 2$, let

$$h = \min_{k \neq l} \frac{n_k n_l}{n_k + n_l} = \min_{k \neq l} \left(\frac{1}{n_k} + \frac{1}{n_l} \right)^{-1}.$$

- (a) For all means with $\max_k \mu_k - \min_k \mu_k \geq \Delta > 0$, the noncentrality of the F test satisfies $\gamma \geq h\Delta^2/\sigma^2$, and there is a configuration with range exactly Δ at which equality holds.
- (b) For fixed $n = \sum_k n_k$, $h \leq n/(2g)$, with equality iff the layout is balanced.

Proof. Part (a) is Exercise (B1), rewritten with h : its bound is $\Delta^2/(\sigma^2(1/n_{(1)} + 1/n_{(2)})) = h\Delta^2/\sigma^2$, since $1/n_k + 1/n_l$ is largest for the two smallest groups. Equality holds when those two groups carry means Δ apart and every other mean equals their weighted average.

For (b): as just noted, $1/n_k + 1/n_l$ is largest for the two smallest groups. Call their sizes $a \leq b$. The other $g - 2$ groups have at least b observations each, so $n \geq a + (g - 1)b$, and

$$\left(\frac{1}{a} + \frac{1}{b} \right) n \geq \left(\frac{1}{a} + \frac{1}{b} \right) (a + (g - 1)b) = g + (g - 1)t + \frac{1}{t},$$

where $t = b/a \geq 1$. Now $(g - 1)t + 1/t - g = ((g - 1)t - 1)(t - 1)/t \geq 0$, because both factors are nonnegative when $t \geq 1$ and $g \geq 2$. Hence $1/a + 1/b \geq 2g/n$, that is, $h \leq n/(2g)$.

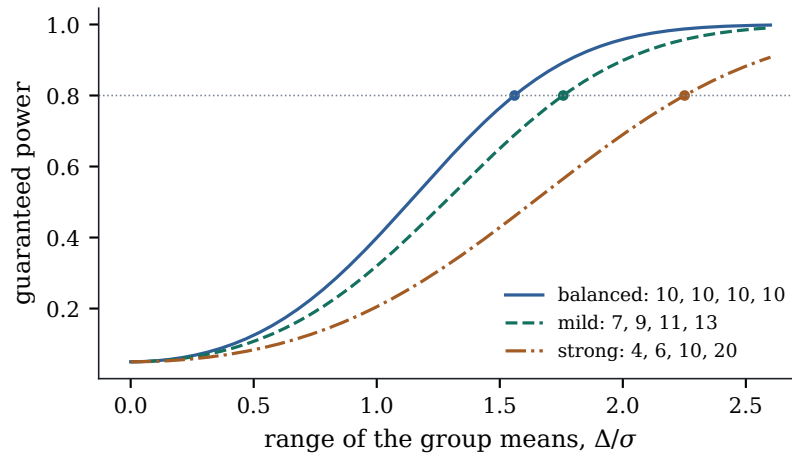


Figure 15.5.1: Guaranteed power of the level-0.05 F test with 40 observations in four groups, against the range of the group means in units of σ , for three allocations. The curves are the power at the least favourable configuration of Proposition 15.5.1; dots mark where each reaches 0.8.

Equality requires $t = 1$ and $n = a + (g - 1)b$, so all groups have the same size. Conversely, a balanced layout has $h = m/2 = n/(2g)$. ■

A departure is hardest to detect when it sits in the least informative pair of groups. The balanced value $h = m/2$ is the bound of Proposition 11.5.2.

Example. Forty observations in four groups

With $n = 40$ and $g = 4$, the guaranteed noncentrality per unit of $(\Delta/\sigma)^2$ is 5.000 for the balanced allocation 10, 10, 10, 10, 3.938 for the mildly unbalanced 7, 9, 11, 13, and 2.400 for 4, 6, 10, 20. At $\Delta = \sigma$ the guaranteed powers of the level-0.05 test are 0.399, 0.320 and 0.205. To guarantee power 0.8, the range of the means must be at least 1.56σ , 1.76σ and 2.25σ respectively (Figure 15.5.1). The mild imbalance gives the same guarantee as a balanced design with $8 \times 3.938 \approx 31.5$ observations, so it wastes a fifth of the experiment.

```
import itertools

import numpy as np
from scipy import stats

def guaranteed_gamma(sizes, delta_over_sigma):
    """Smallest noncentrality over all mean configurations with range at least Delta
    """
    h = min(a * b / (a + b) for a, b in itertools.combinations(sizes, 2))
    return delta_over_sigma ** 2 * h

def power_f(gamma, q, nu, alpha=0.05):
    return stats.ncf.sf(stats.f.ppf(1 - alpha, q, nu), q, nu, gamma)

allocations = {"balanced": [10, 10, 10, 10], "mild": [7, 9, 11, 13], "strong": [4,
    6, 10, 20]}
for name, sizes in allocations.items():
```

```
gam = guaranteed_gamma(sizes, 1.0)
print(f"{name:9s} {sizes}: guaranteed gamma = {gam:.3f},"
      f" guaranteed power at Delta = sigma: {power_f(gam, 3, 36):.3f}")
```

[Run this code online](#)

Balance is optimal for the overall test and for the family of all pairwise comparisons. It is not optimal for every purpose. A single planned contrast is estimated most precisely with $n_k \propto |c_k|$ (Exercise (B2)), and comparisons with a control favour a larger control group (Exercise (B3)). The allocation should follow the questions.

15.5.3 Unequal variances

The one-way model assumes a common variance. Suppose instead that the observations in group k have variance σ_k^2 , with everything else as before. What do the two mean squares then estimate?

Theorem 15.5.2. Expected mean squares under unequal variances

Let the Y_{kj} be uncorrelated with $E(Y_{kj}) = \mu_k$ and $\text{Var}(Y_{kj}) = \sigma_k^2$, and write $\bar{n} = n/g$. Then:

- (a) $E(\text{SSE}) = \sum_k (n_k - 1) \sigma_k^2$, so $E(\text{MSE})$ is the average of the σ_k^2 weighted by $n_k - 1$;
- (b) $E(\text{SSB}) = \sum_k (1 - n_k/n) \sigma_k^2 + \sum_k n_k (\mu_k - \bar{\mu})^2$;
- (c) when the means are equal,

$$E(\text{MSB}) - E(\text{MSE}) = \frac{g(n-1)}{n(g-1)(n-g)} \sum_{k=1}^g (\bar{n} - n_k) \sigma_k^2. \quad (15.5.1)$$

In a balanced layout this is zero, and both mean squares estimate $g^{-1} \sum_k \sigma_k^2$. In an unbalanced layout it is positive when the larger variances belong to the smaller groups and negative when they belong to the larger groups.

Proof. The argument of Theorem 9.6.1 applies with a diagonal covariance. Now $\text{Cov}(\mathbf{Y}) = \mathbf{\Sigma}$, diagonal with σ_k^2 on the observations of group k , and $E(\mathbf{Y}) = \mathbf{\theta}$. By Theorem 2.3.1, $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \mathbf{\Sigma}) + \mathbf{\theta}^\top \mathbf{A} \mathbf{\theta}$. (a) With $\mathbf{A} = \mathbf{I} - \mathbf{M}$, whose diagonal entries in group k are $1 - 1/n_k$, $\text{tr}(\mathbf{A} \mathbf{\Sigma}) = \sum_k n_k (1 - 1/n_k) \sigma_k^2$, and $\mathbf{A} \mathbf{\theta} = \mathbf{0}$. (b) With $\mathbf{A} = \mathbf{M} - \mathbf{P}_0$, the diagonal entries in group k are $1/n_k - 1/n$, so $\text{tr}(\mathbf{A} \mathbf{\Sigma}) = \sum_k (1 - n_k/n) \sigma_k^2$, and $\mathbf{\theta}^\top \mathbf{A} \mathbf{\theta} = \sum_k n_k (\mu_k - \bar{\mu})^2$ as in Theorem 15.3.1. (c) With equal means, multiply the difference of the two expectations by $(g-1)(n-g)$:

$$\begin{aligned} & (n-g) \sum_k (1 - n_k/n) \sigma_k^2 - (g-1) \sum_k (n_k - 1) \sigma_k^2 \\ &= \sum_k (n-1 - g(n-1)n_k/n) \sigma_k^2 = \frac{g(n-1)}{n} \sum_k (\bar{n} - n_k) \sigma_k^2, \end{aligned}$$

where the middle step collects the coefficient of σ_k^2 : $(n-g) - (n-g)n_k/n - (g-1)n_k + (g-1) = n-1 - gn_k + gn_k/n$. Since $\sum_k (\bar{n} - n_k) = 0$, the sum equals $-\sum_k (n_k - \bar{n})(\sigma_k^2 - \bar{\sigma}^2)$ for any constant $\bar{\sigma}^2$, which is negative when n_k and σ_k^2 move together and positive when they move in opposite directions. In a balanced layout every $\bar{n} - n_k$ is zero, and then $E(\text{MSE}) = \sum_k (m-1) \sigma_k^2 / (g(m-1)) = g^{-1} \sum_k \sigma_k^2$.

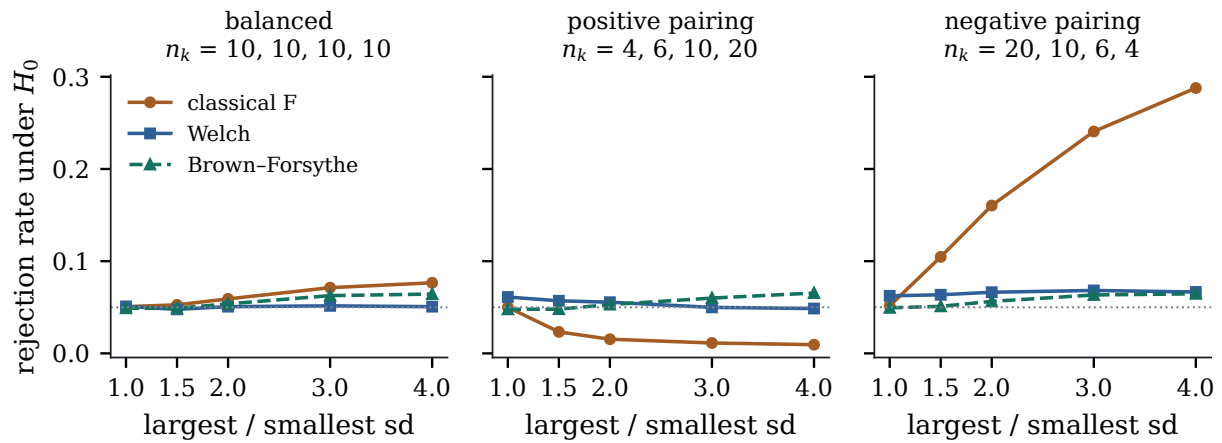


Figure 15.5.2: Rejection rates under equal means of three tests at nominal level 0.05 (dotted), for four normal groups whose standard deviations increase from 1 to the ratio on the horizontal axis. Each point is based on 40000 simulated data sets.

The theorem predicts the behaviour of the F test. In a balanced layout both mean squares estimate the same average variance under the hypothesis, and although the ratio is no longer exactly F -distributed, Box (1954) showed that the effect on the level is modest. In an unbalanced layout MSE leans towards the variances of the large groups and MSB towards those of the small groups. When the small groups are the variable ones the test rejects too often; when the large groups are, it rejects too rarely and loses power.

Example. How far the level moves

Four normal groups have equal means and standard deviations increasing geometrically from 1 to R . Three designs of 40 observations are compared: balanced (10 per group), *positive pairing* (sizes 4, 6, 10, 20, the largest group most variable) and *negative pairing* (sizes 20, 10, 6, 4, the smallest group most variable). For each design and each R , 40000 data sets were simulated and three level-0.05 tests applied: the classical F test and the tests of Welch and of Brown and Forsythe described below (Figure 15.5.2).

At $R = 3$, Eq. 15.5.1 gives expected mean square ratios $E(\text{MSB})/E(\text{MSE})$ of 1.000, 0.559 and 1.923 for the three designs. The simulated levels of the classical test follow: 0.071 when balanced, 0.011 with positive pairing, and 0.241 with negative pairing, almost five times the nominal level. At $R = 4$ the negative pairing gives 0.288. Welch's test stays between 0.048 and 0.068 throughout, and the Brown–Forsythe test between 0.048 and 0.065.

The simulation supports the common rule of thumb that a ratio of about two between the largest and smallest standard deviations does little harm, but only with balance: at $R = 2$ the balanced level is 0.059, while the two unbalanced designs give 0.015 and 0.160.

15.5.4 Tests that allow unequal variances

If the variances σ_k^2 were known, weighted least squares (Section 6.9) would compare the group means with their precision-weighted average $\tilde{\mu} = \sum_k w_k \bar{Y}_k / \sum_k w_k$, $w_k = n_k / \sigma_k^2$, and $\sum_k w_k (\bar{Y}_k - \tilde{\mu})^2$ would have an exact $\chi^2(g - 1)$ law under equal means. In practice the σ_k^2 are replaced by

the s_k^2 and the reference distribution is approximated. We describe the two approximations in common use without deriving them; [Chapter 21](#) treats heteroscedasticity in general.

Welch's test. Welch (1951) replaced σ_k^2 by s_k^2 in the weights, $w_k = n_k/s_k^2$, and divided the weighted mean square by a correction factor:

$$F_W = \frac{\sum_k w_k (\bar{y}_k - \bar{\mu})^2 / (g - 1)}{1 + \frac{2(g - 2)}{g^2 - 1} \sum_k h_k}, \quad h_k = \frac{(1 - w_k / \sum_l w_l)^2}{n_k - 1},$$

referred to $F(g - 1, \nu_W)$ with $\nu_W = (g^2 - 1) / (3 \sum_k h_k)$. The correction and the degrees of freedom come from an approximation to the statistic's null distribution that is accurate to first order in the quantities $1/(n_k - 1)$. For two groups F_W is the square of Welch's two-sample t statistic with Satterthwaite's degrees of freedom ([Exercise \(B1\)](#)).

The Brown–Forsythe test. Brown and Forsythe (1974a) kept the unweighted numerator and changed the denominator. By [Theorem 15.5.2\(b\)](#), under equal means SSB has expectation $\sum_k (1 - n_k/n) \sigma_k^2$, whatever the variances. Replacing σ_k^2 by s_k^2 gives an unbiased estimate of that expectation, and the statistic

$$F_{BF} = \frac{\text{SSB}}{\sum_k (1 - n_k/n) s_k^2}$$

compares SSB with an estimate of what it should be under the hypothesis. It is referred to $F(g - 1, f)$, where $1/f = \sum_k c_k^2 / (n_k - 1)$ with $c_k = (1 - n_k/n) s_k^2 / \sum_l (1 - n_l/n) s_l^2$, Satterthwaite's approximation to the degrees of freedom of a linear combination of independent variance estimates. In a balanced layout the denominator is $(g - 1)s^2$, so $F_{BF} = F$ and only the degrees of freedom change.

Individual contrasts need the same care, since $s\sqrt{v_c}$ estimates the wrong variance when the σ_k^2 differ, even in a balanced layout ([Exercise \(C2\)](#)). The **Welch–Satterthwaite** interval uses

$$\widehat{\text{Var}}(\hat{\psi}) = \sum_k \frac{c_k^2 s_k^2}{n_k}, \quad \nu = \frac{(\sum_k c_k^2 s_k^2 / n_k)^2}{\sum_k (c_k^2 s_k^2 / n_k)^2 / (n_k - 1)},$$

and the multiplier $t_{\nu, \alpha/2}$ ([Exercise \(B3\)](#)), adjusted as in [Chapter 13](#) for simultaneous inference.

Example. An unbalanced trial with unequal variances

A repair mortar was made with four admixture formulations, and its setting time was measured in minutes. The standard formulation A was tested most often, and the newer formulations least (simulated data, rounded to 0.1 minute):

Formulation	n_k	Setting times (min)	Mean	s_k
A	12	18.9, 20.2, 20.6, 21.2, 20.3, 19.0, 20.2, 19.0, 20.5, 20.6, 20.1, 20.3	20.075	0.73
B	9	22.6, 21.3, 22.2, 22.8, 19.3, 20.7, 21.5, 20.8, 18.8	21.111	1.38
C	6	20.7, 18.2, 18.6, 23.6, 21.3, 22.6	20.833	2.14
D	4	20.1, 22.6, 28.3, 25.3	24.075	3.53

The standard deviations grow as the group sizes shrink, the negative pairing of [Example \(How far the level moves\)](#). The classical analysis gives $F = 5.34$ on $(3, 27)$ degrees of freedom and $p = 0.0051$, apparently strong evidence, driven by the high mean of D. But $\text{MSE} = 3.015$ gives the variance of D weight $3/27$, while the null expectation of MSB in [Theorem 15.5.2\(b\)](#) gives it weight $(1 - 4/31)/3 \approx 0.29$; estimated with the s_k^2 , that expectation is 5.407, far above MSE. Welch's test gives $F_W = 2.64$ on $(3, 8.3)$ degrees of freedom and $p = 0.119$, and the Brown–Forsythe test gives $F_{BF} = 2.98$ on $(3, 6.2)$ degrees of freedom and $p = 0.115$. The difference between D and A is 4.00 minutes. With the pooled standard error 1.003 its 95% interval is $[1.94, 6.06]$; with the Welch–Satterthwaite standard error 1.776 on 3.1 degrees of freedom it is $[-1.56, 9.56]$. Four observations with a standard deviation of 3.5 minutes cannot establish a difference of four minutes.

The weighted and unweighted averages of the group means are 21.039 and 21.524, so the side conditions $\sum_k n_k \alpha_k = 0$ and $\sum_k \alpha_k = 0$ of [Section 15.1](#) (Theorem 8.4.2) define overall means that differ by half a minute here.

```
import numpy as np
from scipy import stats
from statsmodels.stats.oneway import anova_oneway

labels = ["A", "B", "C", "D"]
n_k = np.array([12, 9, 6, 4])
rng = np.random.default_rng(1520)
samples = [np.round(rng.normal(mu, sd, k), 1)
            for mu, sd, k in zip([20.0, 20.6, 21.0, 22.2], [0.8, 1.2, 2.0, 3.0], n_k)
          ]

g, n = len(samples), n_k.sum()
means = np.array([x.mean() for x in samples])
s2_k = np.array([x.var(ddof=1) for x in samples]) # group variances
s2 = np.sum((n_k - 1) * s2_k) / (n - g)          # pooled within-group
variance
grand = means @ n_k / n
ss_b = np.sum(n_k * (means - grand) ** 2)
F = ss_b / (g - 1) / s2
print("means:", means.round(3), " sds:", np.sqrt(s2_k).round(2))
print(f"classical F = {F:.2f}, p = {stats.f.sf(F, g - 1, n - g):.4f}")
# null expectation of MS_between if the variances really are s2_k (the theorem of
  this section)
print(f"MS_within = {s2:.3f}; estimated null E(MS_between) = "
      f"{np.sum((1 - n_k / n) * s2_k) / (g - 1):.3f}")

welch = anova_oneway(samples, use_var="unequal")
bf = anova_oneway(samples, use_var="bf")
print(f"Welch: F = {welch.statistic:.2f} on ({g - 1}, {welch.df_denom:.1f}) df, p = "
      f"{welch.pvalue:.3f}")
print(f"Brown-Forsythe: F = {bf.statistic:.2f} on ({g - 1}, {bf.df2[1]:.1f}) df, p = "
      f"{bf.pvalue2:.3f}")
lev = stats.levene(*samples, center="median") # dispersion test on |y -
median/
print(f"variance test on absolute deviations from medians: p = {lev.pvalue:.4f}")
```

[Run this code online](#)

(By default statsmodels refines the numerator degrees of freedom of the Brown–Forsythe test; pvalue2 is the original version.)

The simulation of [Example \(How far the level moves\)](#) used the same formulas, applied to group means and variances drawn directly from their sampling distributions:

```
import numpy as np
from scipy import stats

def three_tests(ybar, s2, n_k):
    """p-values of the classical F, Welch and Brown-Forsythe tests; arrays of shape
    (reps, g)."""
    g, n = n_k.size, n_k.sum()
    grand = ybar @ n_k / n
    ss_b = ((ybar - grand[:, None]) ** 2) @ n_k
    ss_w = s2 @ (n_k - 1)
    p_f = stats.f.sf(ss_b / (g - 1) / (ss_w / (n - g)), g - 1, n - g)
    # Welch (1951): weights n_k / s_k^2 and a Satterthwaite-type denominator df
    w = n_k / s2
    mu_w = np.sum(w * ybar, axis=1) / w.sum(axis=1)
    h = (1 - w / w.sum(axis=1)[:, None]) ** 2 / (n_k - 1)
    a = np.sum(w * (ybar - mu_w[:, None]) ** 2, axis=1) / (g - 1)
    b = 1 + 2 * (g - 2) / (g ** 2 - 1) * h.sum(axis=1)
    p_w = stats.f.sf(a / b, g - 1, (g ** 2 - 1) / (3 * h.sum(axis=1)))
    # Brown-Forsythe (1974): SS_between over its null expectation under unequal
    variances
    d = (1 - n_k / n) * s2
    c = d / d.sum(axis=1)[:, None]
    p_bf = stats.f.sf(ss_b / d.sum(axis=1), g - 1, 1 / np.sum(c ** 2 / (n_k - 1),
axis=1))
    return p_f, p_w, p_bf

def size(n_k, sds, reps=40000, seed=0, alpha=0.05):
    """Rejection rates under equal means, from group means and variances drawn
    exactly."""
    rng = np.random.default_rng(seed)
    n_k, sds = np.asarray(n_k, float), np.asarray(sds, float)
    ybar = rng.normal(0, sds / np.sqrt(n_k), (reps, n_k.size))
    s2 = sds ** 2 * rng.chisquare(n_k - 1, (reps, n_k.size)) / (n_k - 1)
    return [np.mean(p < alpha) for p in three_tests(ybar, s2, n_k)]

designs = {"balanced": [10, 10, 10, 10],
          "positive pairing": [4, 6, 10, 20],          # big groups get the big
          variances
          "negative pairing": [20, 10, 6, 4]}          # small groups get the big
          variances
ratios = [1, 1.5, 2, 3, 4]
res = {}
for name, sizes in designs.items():                  # sds grow geometrically from 1
    to R
    res[name] = np.array([size(sizes, R ** (np.arange(4) / 3), seed=int(10 * R)) for
    R in ratios])
for R, row in zip(ratios, res[name]):
    print(f"{name:17s} R = {R:3.1f}: F, Welch, BF sizes = {np.round(row, 3)}")
```

[Run this code online](#)

Robustness has a small price. With equal variances and a group of only four observations, Welch's test rejects slightly too often, 0.061 and 0.062 in the two unbalanced designs. The practical advice is simple: with unequal group sizes, use a heteroscedasticity-robust test unless there is good reason to believe the variances are equal; with equal sizes the classical test is

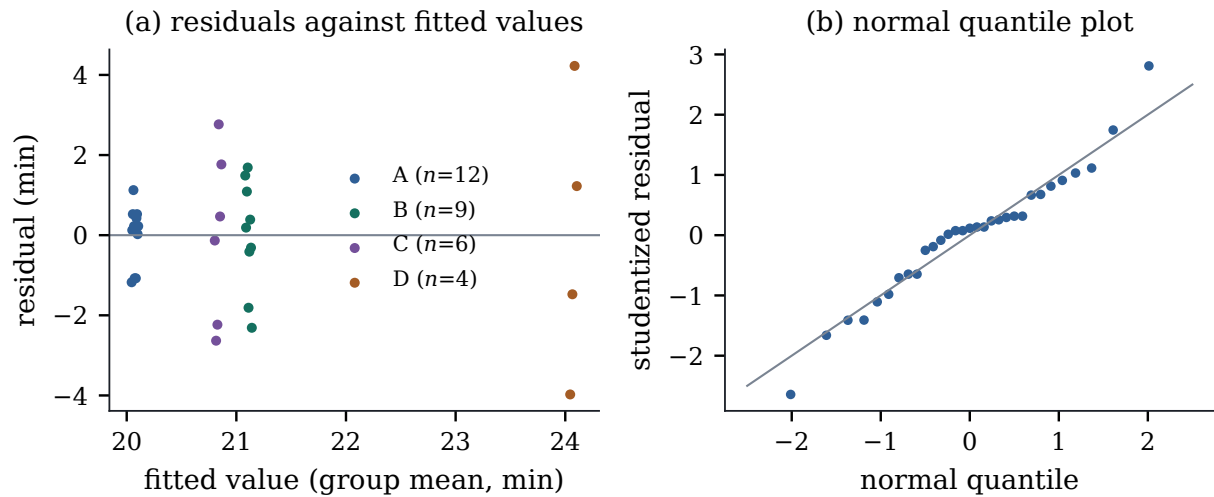


Figure 15.5.3: Residual plots for the mortar trial. (a) Residuals against fitted values: the spread grows from formulation A to formulation D. (b) Normal quantile plot of the studentized residuals: the curvature at both ends reflects the mixture of groups with different variances, not a heavy-tailed error distribution.

acceptable, although individual contrasts still need care.

15.5.5 Checking the assumptions

Independence is secured by the design, above all by randomization, and cannot be checked well from the data. Correlation within groups, which arises when the observations in a group share a batch, an operator or a day, inflates the F statistic badly ([Exercise \(C1\)](#)). A common variance and normality are checked with the residuals $e_{kj} = y_{kj} - \bar{y}_k$.

The leverage of an observation in group k is $1/n_k$ (Proposition 6.8.1), so $\text{Var}(e_{kj}) = \sigma^2(1 - 1/n_k)$. Residuals from small groups are shrunk towards zero, so they are compared on the common scale $r_{kj} = e_{kj}/(s\sqrt{1 - 1/n_k})$. A plot of residuals against the group means shows whether the spread differs between groups, and a spread growing with the mean suggests a transformation ([Chapter 22](#)). A normal quantile plot of the studentized residuals shows skewness and heavy tails. [Chapter 20](#) develops residual diagnostics in general.

[Figure 15.5.3](#) shows the two plots for the mortar trial. The fan shape in panel (a) is unmistakable. The classical tests of equal variances, which compare the logarithms of the s_k^2 , are so sensitive to nonnormality that a rejection says as much about the tails as about the variances. Levene (1960) proposed instead to apply the one-way F test to the absolute deviations $|y_{kj} - \bar{y}_k|$, which are larger on average in a more variable group; Brown and Forsythe (1974b) used deviations from the group *medians*, which behave better for skewed data. For the mortar trial that version gives $p = 0.0019$. Using such a test as a gate for the classical F test is not recommended, since it has little power in small groups, exactly where unequal variances do the most harm. The analysis should be chosen in advance, from the design and from what is known about the response.

Nonnormality is usually the least serious problem for tests about means: the level of the F test is fairly robust to it, more so in balanced designs (Proposition 19.3.5), though heavy tails cost power by inflating MSE. Transformations ([Chapter 22](#)) and resampling ([Chapter 23](#)), for example the Kruskal–Wallis permutation test on ranks, are the usual remedies.

15.5.6 Exercises

15.5.6.1 A. Check your understanding

Exercise. A1

For the balanced design with $m = 10$ and variances 1, 2, 4, 8, compute $E(\text{MSB})$ under equal means and $E(\text{MSE})$ from [Theorem 15.5.2](#). Repeat for the sizes 4, 6, 10, 20 and for 20, 10, 6, 4.

Solution. Balanced: both equal $(1 + 2 + 4 + 8)/4 = 3.75$. Sizes 4, 6, 10, 20 ($n = 40$): $E(\text{SSB}) = 0.9 + 0.85 \cdot 2 + 0.75 \cdot 4 + 0.5 \cdot 8 = 9.6$, so $E(\text{MSB}) = 3.2$; $E(\text{SSE}) = 3 + 10 + 36 + 152 = 201$, so $E(\text{MSE}) = 201/36 \approx 5.58$. The test is conservative. Sizes 20, 10, 6, 4: $E(\text{SSB}) = 0.5 + 0.75 \cdot 2 + 0.85 \cdot 4 + 0.9 \cdot 8 = 12.6$, so $E(\text{MSB}) = 4.2$; $E(\text{SSE}) = 19 + 18 + 20 + 24 = 81$, so $E(\text{MSE}) = 2.25$. The test is liberal.

Exercise. A2

Show that the hat matrix of the one-way model has diagonal entries $1/n_k$ and that $\text{Var}(e_{kj}) = \sigma^2(1 - 1/n_k)$. What is the residual of an observation that is alone in its group, and why is it useless for checking the model?

Solution. The hat matrix is $\mathbf{M}$ of Eq. 15.1.1, whose diagonal entry for an observation of group k is $1/n_k$. By Proposition 5.5.3, $\text{Cov}(\hat{\boldsymbol{\epsilon}}) = \sigma^2(\mathbf{I} - \mathbf{M})$, so $\text{Var}(e_{kj}) = \sigma^2(1 - 1/n_k)$. An observation alone in its group has leverage one, its fitted value is itself, and its residual is identically zero: it carries no information about the variance or about the model.

15.5.6.2 B. Practice

Exercise. B1

Show that for $g = 2$ the Welch statistic F_W equals t_W^2 , where $t_W = (\bar{y}_1 - \bar{y}_2)/\sqrt{s_1^2/n_1 + s_2^2/n_2}$, and that ν_W is Satterthwaite's $(s_1^2/n_1 + s_2^2/n_2)^2 / [(s_1^2/n_1)^2/(n_1 - 1) + (s_2^2/n_2)^2/(n_2 - 1)]$.

Solution. With $g = 2$ the correction factor is 1. Write $v_k = s_k^2/n_k = 1/w_k$ and $W = w_1 + w_2$. The weighted sum of squares of two numbers about their weighted mean is $w_1 w_2 (\bar{y}_1 - \bar{y}_2)^2 / W = (\bar{y}_1 - \bar{y}_2)^2 / (v_1 + v_2)$, which is t_W^2 . Next $1 - w_1/W = w_2/W = v_1/(v_1 + v_2)$, so $h_1 = v_1^2 / \{(v_1 + v_2)^2(n_1 - 1)\}$ and similarly for h_2 . Then $\nu_W = 3 / (3 \sum_k h_k) = (v_1 + v_2)^2 / \{v_1^2/(n_1 - 1) + v_2^2/(n_2 - 1)\}$.

Exercise. B2

Show that for $g = 2$ the Brown–Forsythe statistic also equals t_W^2 , and that its degrees of freedom f equal ν_W . So the two robust tests coincide for two groups and differ only for three or more.

Solution. With two groups, $SSB = n_1 n_2 (\bar{y}_1 - \bar{y}_2)^2 / n$, and the denominator is $(n_2 s_1^2 + n_1 s_2^2) / n$. Their ratio is $(\bar{y}_1 - \bar{y}_2)^2 / (s_1^2 / n_1 + s_2^2 / n_2) = t_W^2$. The weights are $c_1 = (n_2 s_1^2 / n) / \{(n_2 s_1^2 + n_1 s_2^2) / n\} = v_1 / (v_1 + v_2)$ and $c_2 = v_2 / (v_1 + v_2)$, so $1/f = \sum_k c_k^2 / (n_k - 1) = 1/\nu_W$ by the previous exercise.

Exercise. B3

Let $\hat{V} = \sum_k c_k^2 s_k^2 / n_k$ with independent $s_k^2 \sim \sigma_k^2 \chi^2(n_k - 1) / (n_k - 1)$ under normality. Show that $E \hat{V} = V = \text{Var}(\hat{\psi})$ and $\text{Var} \hat{V} = 2 \sum_k (c_k^2 \sigma_k^2 / n_k)^2 / (n_k - 1)$. Choose ν so that $\nu \hat{V} / V$ has the mean and variance of a $\chi^2(\nu)$ variable, and show that replacing σ_k^2 by s_k^2 in the answer gives the degrees of freedom of the Welch–Satterthwaite interval.

Solution. Since $E s_k^2 = \sigma_k^2$, $E \hat{V} = \sum_k c_k^2 \sigma_k^2 / n_k = V$, which is $\text{Var}(\hat{\psi})$ because the group means are independent with variances σ_k^2 / n_k . $\text{Var}(s_k^2) = 2\sigma_k^4 / (n_k - 1)$, and the terms are independent, which gives $\text{Var} \hat{V}$. A $\chi^2(\nu)$ variable has mean ν and variance 2ν . The mean of $\nu \hat{V} / V$ is ν for every ν , and its variance is $\nu^2 \text{Var}(\hat{V}) / V^2$. Setting this equal to 2ν gives $\nu = 2V^2 / \text{Var} \hat{V} = V^2 / \sum_k (c_k^2 \sigma_k^2 / n_k)^2 / (n_k - 1)$. With s_k^2 for σ_k^2 this is the formula in the text.

Exercise. B4

For $n = 30$ observations in five groups, compare the quantity h of [Proposition 15.5.1](#) for the balanced allocation and for the allocation 10, 5, 5, 5, 5 suggested by Exercise (B3) for comparisons with a control. What does each allocation optimize?

Solution. Balanced: $h = 6/2 = 3$. For 10, 5, 5, 5, 5, the least informative pair is two treatments of size 5, with $h = 25/10 = 2.5$. The guaranteed noncentrality of the overall test is $3(\Delta/\sigma)^2$ against $2.5(\Delta/\sigma)^2$. But each comparison with the control has variance $\sigma^2(1/10 + 1/5) = 0.3\sigma^2$ instead of $\sigma^2/3$. Balance protects the overall test and the comparisons among treatments; the unbalanced allocation sharpens the comparisons with the control.

15.5.6.3 C. Going deeper

Exercise. C1

In a balanced layout, suppose that under equal means the errors within a group are equicorrelated, $\text{Cov}(\varepsilon_{kj}, \varepsilon_{kj'}) = \rho\sigma^2$ for $j \neq j'$ with $\rho \geq 0$, and independent between groups. Show that $E(\text{MSE}) = \sigma^2(1 - \rho)$ and $E(\text{MSB}) = \sigma^2(1 - \rho + m\rho)$. What happens to the level of the F test with $m = 10$ and $\rho = 0.1$?

Solution. The covariance matrix is $\sigma^2\{(1 - \rho)\mathbf{I} + \rho\mathbf{Z}\mathbf{Z}^\top\}$, the random-effects form with $\sigma_a^2 = \rho\sigma^2$ and error variance $(1 - \rho)\sigma^2$. By Exercise (C1) with $n_0 = m$, $E(\text{MSE}) = (1 - \rho)\sigma^2$ and $E(\text{MSB}) = (1 - \rho)\sigma^2 + m\rho\sigma^2$. Under normality $\text{MSB} / E(\text{MSB})$ and $\text{MSE} / E(\text{MSE})$ are independent scaled chi-squared variables, so F is $1 + m\rho / (1 - \rho)$ times an $F(g - 1, n - g)$

variable. With $m = 10$ and $\rho = 0.1$ the factor is $1 + 1/0.9 \approx 2.1$, and a nominal 5% test rejects far more often: with $g = 4$ groups its level is 0.271. A small intraclass correlation does great damage when groups are large.

Exercise. C2

In a balanced layout with variances σ_k^2 , show that the pooled variance estimate of the pairwise difference $\bar{Y}_k - \bar{Y}_l$, $2s^2/m$, has expectation $2\bar{\sigma}^2/m$ with $\bar{\sigma}^2 = g^{-1} \sum_i \sigma_i^2$, while the true variance is $(\sigma_k^2 + \sigma_l^2)/m$. Conclude that balance protects the overall test but not individual comparisons, and find the ratio of true to estimated variance for the two most variable groups when the variances are 1, 2, 4, 8.

Solution. By [Theorem 15.5.2\(c\)](#), in a balanced layout $E(s^2) = \bar{\sigma}^2$, so $E(2s^2/m) = 2\bar{\sigma}^2/m$, and the true variance is $\sigma_k^2/m + \sigma_l^2/m$ because the means are independent. For the groups with variances 4 and 8, the true variance is $12/m$ and the pooled estimate targets $2 \cdot 3.75/m = 7.5/m$, a ratio of 1.6. Intervals for that difference built from s^2 are too short by a factor of about $\sqrt{1.6} \approx 1.26$, and intervals for the difference of the two least variable groups are too long.